File Handling in C language

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File handling in C is about persisting data to a secondary storage (disk) and retrieving it later using standard IO functions from <stdio.h>
It revolves around the **FILE** abstraction and a set of APIs for opening, reading, writing and managing files.

What is a File in C?
A file is a named collection of data stored on disk. C treat files as stream of bytes.

Types of Files:
1. Text files ->human readable(e.g., .py, .txt, .c, .cpp etc.)
2. Binary files -> raw bytes(not comfortable to read by humans)
(e.g., .dat, .bin, .png, .mp4, .mp3, .doc, .pdf, etc.)

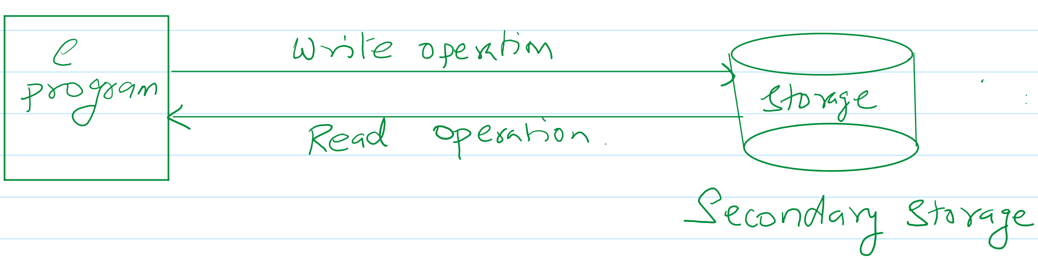
File Pointer
C uses a pointer of type FILE to work with files.

```
FILE *fp;
```

- It stores meta data (buffer position mode, etc.)
- Must be associated with real file via fopen()

Opening a file using fopen() function:

```
fp = fopen("filename", "mode");
```



File Mode	Meaning
"r"	Read(file must exist)
"w"	Write(create/overwrite)
"a"	Append(if file is existing then add the data, after existing data, otherwise it will create a new file)
"r+"	Read + Write
"w+"	Read + Write(overwrite)
"a+"	Read + Append

```
FILE *fp = fopen("data.txt", "r");
```

Closing the file

```
fclose(fp);
```

Always close flies to:

- Flush buffers
- Release resources

How to create a file by using C-language?

```
#include<stdio.h>
#include<string.h>
#include<stdlib.h>
#define SIZE 200
int main(int argc, char const *argv[])
{
    char fileName[SIZE];
    printf("Enter file name: ");
    scanf("%s", fileName);
    char path[SIZE] = "../Files/";
    strcat(path, fileName);
    FILE *fp = fopen(path, "w");
    if(fp == NULL){
        printf("File not created!");
        exit(1);
    }
```

is common code in every file handing program.

```
    char name[SIZE];
    printf("Enter your name: ");
    scanf("%s", name);

    fprintf(fp, "Name: %s\n", name);
```

function to write data into the file.

file pointer

data written in the file

```
    return 0;
}
```

Read data from file

```
char buffer[SIZE];
fscanf(fp, "%s", buffer);
```

Example:

```
#include<stdio.h>
#include<string.h>
#include<stdlib.h>
#define SIZE 200
```

```c
int main(int argc, char const *argv[])
{
    char fileName[SIZE];
    printf("Enter file name: ");
    scanf("%s", fileName);
    char path[SIZE] = "../Files/";
    strcat(path, fileName);
    FILE *fp = fopen(path, "r");
    if(fp == NULL){
        printf("File not found on the specified path!");
        exit(1);
    }
    char name[SIZE];
    // fscanf(fp, "%s", name);
    printf("\nData from file:\n");
    while(fgets(name, SIZE, fp)!= NULL){
        printf("%s", name);
    }
    return 0;
}
```

Output:

```
Enter file name: name.txt

Data from file:
Name: Shiv
Address: Garia
City: Kolkata
```

```c
#include<stdio.h>
#include<string.h>
#include<stdlib.h>
#define SIZE 200
int main(int argc, char const *argv[])
{
    char fileName[SIZE];
    printf("Enter file name: ");
    scanf("%s", fileName);
    char path[SIZE] = "../Files/";
    strcat(path, fileName);
    FILE *fp = fopen(path, "a");
    if(fp == NULL){
        printf("File not created!");
        exit(1);
    }
    char name[SIZE];
    printf("Enter your name: ");
    scanf("%s", name);
    strcat(name, "\n");
    fputs(name, fp);
    return 0;
}
```